Deterministic Optimization

Further Development of Simplex Method

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Summarize Simplex Method

Simplex Method in a Nutshell

Learning Objectives for this module

- Explore some advanced features of simplex method, including how to handle degeneracy, and how to find an initial basic feasible solution.

Simplex Method Recap

- Simplex method starts from a Basic Feasible Solution (BFS)
- Check reduced costs of all nonbasic variables
    - If all reduced costs are nonnegative, then the current BFS is optimal. Simplex method terminates.
    - Otherwise, pick a nonbasic variable with negative reduced cost to enter the basis
- Compute the direction pointing from the current BFS to the adjacent BFS determined by the nonbasic variable that enters the basis
    - If the direction vector is nonnegative, then the LP has unbounded optimum. Simplex method terminates.
    - Otherwise, do min-ratio test to find a basic variable to leave basis. Continue simplex method from the new BFS.

Summary of Simplex Method

1. We start the simplex method with a basic feasible solution $\boldsymbol{x}$ and the associated basis consisting of the basis matrix $\boldsymbol{B} = [\boldsymbol{A}_{B(1)}, \ldots, \boldsymbol{A}_{B(m)}]$, where $\boldsymbol{A}_{B(i)}$'s are the basic columns of $\boldsymbol{A}$.

2. Compute the reduced cost $\bar{c}_j = c_j - \boldsymbol{c}_B^\top \boldsymbol{B}^{-1} \boldsymbol{A}_j$ for each nonbasic variable x_j. There are two possibilities:

   (2.1) If all the reduced costs are nonnegative, the current BFS is optimal, and the algorithm terminates;

   (2.2) Otherwise, select some j with $\bar{c}_j < 0$, so x_j *enters the basis.*

3. Compute $\boldsymbol{d}_B = -\boldsymbol{B}^{-1}\boldsymbol{A}_j$. There are two possibilities:

   (3.1) If $\boldsymbol{d}_B \geq \boldsymbol{0}$, we know the optimal cost is $-\infty$, and the algorithm terminates.

   (3.2) Otherwise, if some entry of $\boldsymbol{d}_B$ is negative, continue to the next step.

4. If some entry of $\boldsymbol{d}_B$ is negative, compute the stepsize θ^* by the following *min-ratio test*:

$$\theta^* = \min_{\{i=1,\ldots,m \mid d_{B(i)}<0\}} \frac{x_{B(i)}}{-d_{B(i)}}.$$

   Suppose the index $B(l)$ achieves the minimum: $x_{B(l)}$ *exits the basis.*

5. Form the new basis matrix $\overline{\boldsymbol{B}}$ by replacing $\boldsymbol{A}_{B(l)}$ column with $\boldsymbol{A}_j$. The new BFS $\boldsymbol{y}$ has basic variable part $y_{B(i)} = x_{B(i)} + \theta^* d_{B(i)}$ and $y_j = \theta^*$. Start a new iteration.

Correctness of Simplex Method

Theorem 1 (Correctness of the Simplex Method). *If every basic feasible solution of a standard form linear program is nondegenerate, then the simplex method always terminates in a finite number of steps, either*

1. *finds an optimal solution* $\boldsymbol{x}_B = \boldsymbol{B}^{-1}\boldsymbol{b}$ *and* $\boldsymbol{x}_N = \mathbf{0}$, *with the associated optimal basis matrix* $\boldsymbol{B}$*;*

2. *or finds a direction* $\mathbf{d} = (\boldsymbol{d}_B, \boldsymbol{d}_N)$ *such that* $\boldsymbol{A}\boldsymbol{d} = \mathbf{0}$, $\boldsymbol{d}_B \geq \mathbf{0}$, $\boldsymbol{d}_N = (0, \ldots, 0, 1, 0, \ldots, 0)^T$ *with* j*-th element being* 1, $\mathbf{c}^T\boldsymbol{d} < 0$, *and the optimal cost is* $-\infty$.

Recall: a BFS is <u>nondegenerate</u> in standard form LP if all basic variables are positive.

Further Questions

- How to find an initial BFS to start the simplex method?
- What if the LP is infeasible? How to detect infeasibility?
- What if some BFS’s are degenerate? How does degeneracy affect simplex method?
- It turns out that these questions are nontrivial questions. We will address them in the coming lectures.

Summary

- We learned:
  - We summarized simplex method
  - We pointed out some key questions that the basic simplex method does not address